

Month 4 – Projects + DevOps: Detailed Exercises (Weeks 13-16, Days 85-112)

MERN MASTERY · 1 CR PRODUCT PREP · HARSH VARDHAN GUPTA



Study window: 9 Nov 2026 – 6 Dec 2026 (Days 85-112)



Your slots (IST): Mon-Fri **7:30-8:30pm** & **9:00-10:00pm** · Sat-Sun **9:00am-12:00pm**, **3:00-5:30pm**, **6:00-8:30pm**



Calendar reminders: harsh.gupta@getreelax.com – import `calendar/mern-mastery-study.ics`

Roadmap: [ROADMAP.md](#) · Theory: [07-auth-devops-complete.md](#) · Index: [week-13-16-exercises.md](#)

WEEKS 13-14 – TaskFlow Project (Days 85-98)

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Flagship MERN resume project: team task management with real-time updates.

Stack: Next.js + Node/Express + MongoDB + Redis + TypeScript + Docker Compose + Socket.io

Day 85 – Mon: Project Setup

Problem statement

Initialize monorepo or split repos: `taskflow-web`, `taskflow-api`. ESLint, Prettier, TS strict, shared types package optional.

Learning objective

Production-grade project scaffold.

Prerequisites

- Months 1–3 exercises complete

Step-by-step approach

- Create GitHub repo `taskflow`; branch `main`.
- API: Express TS template from Month 3; folder structure `routes/controllers/services/models`.
- Web: `create-next-app` with App Router.
- Root `docker-compose.yml` stub (mongo, redis).
- `.env.example` for both apps; document variables in README.
- Husky pre-commit lint optional.
- First commit: "chore: initial monorepo scaffold".

Hints

- Share `User`, `Task` types via package or duplicated initially.
- Use consistent API base URL env `NEXT_PUBLIC_API_URL`.
- README architecture diagram placeholder.

Exact Answer

- Monorepo with `taskflow-web` (Next.js) + `taskflow-api` (Express TS)

- Docker Compose stub for mongo + redis; both apps start locally

Complete Solution

```
taskflow/
├── docker-compose.yml    # mongo, redis
├── taskflow-api/         # Express + TS + Mongoose
├── taskflow-web/         # Next.js App Router
├── .env.example
└── README.md             # architecture diagram
```

Expected output / acceptance criteria

- Both apps start locally; health check passes; CI-ready structure.

Where to practice

- Local: `mern-mastery/taskflow/`
- Docker: <https://docs.docker.com/compose/>

Common mistakes

- Committing real `.env` secrets.
- CORS not planned early.

80 LPA Interview Twist

Netflix/Meta use Redis for rate limiting, pub/sub, and session store — expect distributed Redis cluster and failover discussion.

Interview connection

Walkthrough flagship project architecture — recruiters focus here.

Time estimate: 2 hours (weekday)

Day 86 – Tue: Database Models

Problem statement

Mongoose models: User, Team, Project, Task, Comment, Invite.

Step-by-step approach

1. **User:** email, passwordHash, name, avatarUrl optional.
2. **Team:** name, ownerId, memberIds[], roles map or Member subdoc.
3. **Project:** teamId, name, description, archived flag.
4. **Task:** projectId, title, description, status enum, priority, assigneeId, dueDate.

5. **Comment:** taskId, authorId, body, timestamps.
6. Indexes: task by projectId+status; user email unique.
7. Seed script with demo team/project/tasks.

Exact Answer

- Models: User, Team, Project, Task, Comment, Invite with proper refs and indexes
- Seed script creates demo team with tasks

Complete Solution

```
const taskSchema = new Schema({
  projectId: { type: Schema.Types.ObjectId, ref: "Project", required: true, index: true },
  title: String,
  status: { type: String, enum: ["todo", "in_progress", "done"], default: "todo" },
  priority: { type: String, enum: ["low", "medium", "high"] },
  assigneeId: { type: Schema.Types.ObjectId, ref: "User" },
}, { timestamps: true });
taskSchema.index({ projectId: 1, status: 1 });
```

Expected output / acceptance criteria

- Seed creates sample data; relationships populate correctly.

80 LPA Interview Twist

Meta/Google security rounds ask about OWASP top 10, token rotation on reuse detection, and zero-trust architecture — auth skeleton must scale to microservices.

Time estimate: 2 hours

Day 87 – Wed: Auth Integration

Problem statement

Wire Month 3 JWT auth into TaskFlow: register, login, refresh, protected routes.

Step-by-step approach

1. Mount auth routes under `/api/auth`.
2. User registration creates default personal team optional.
3. Access token 15m; refresh httpOnly cookie.
4. Next.js: API route proxy or direct cookie handling strategy documented.
5. Middleware `authenticate` on all task routes.

Exact Answer

- Follow acceptance criteria in Expected output section
- Implementation matches step-by-step approach requirements

Complete Solution

```
// prisma/schema.prisma + migrate
npx prisma migrate dev --name init
npx prisma generate
const users = await prisma.user.findMany({ include: { orders: true } });
```

80 LPA Interview Twist

Meta/Google security rounds ask about OWASP top 10, token rotation on reuse detection, and zero-trust architecture — auth skeleton must scale to microservices.

Time estimate: 2 hours

Day 88 – Thu: Teams + Invite API

Problem statement

Create team, invite by email token, accept invite, list members, roles owner/member/viewer.

Step-by-step approach

1. POST `/teams` — creator becomes owner.
2. POST `/teams/:id/invites` — generate token, expiry 7d (email stub log).
3. POST `/invites/accept` — add member with role.
4. GET `/teams/:id/members` — RBAC: viewer can read.
5. PATCH member role — owner only.

Exact Answer

- Follow acceptance criteria in Expected output section
- Implementation matches step-by-step approach requirements

Complete Solution

```
describe("calculateTotal", () => {
  it("sums cart items", () => {
    expect(calculateTotal([{ price: 10, qty: 2 }, { price: 5, qty: 1 }])).toBe(25)
  });
});
```

80 LPA Interview Twist

At Google/Meta 80 LPA level, expect follow-up questions on scale (10× traffic), failure modes, and production tradeoffs — not just happy-path implementation.

Time estimate: 2 hours

Day 89 – Fri: LeetCode LC 253

Problem statement

Meeting Rooms II — min rooms for overlapping intervals.

Step-by-step approach

1. Sort by start time.
2. Min-heap on end times; if start $\geq$ min end pop.
3. Heap size = rooms needed.

Exact Answer

- Optimal solution with stated time/space complexity
- Passes all LeetCode test cases including edge cases

Complete Solution

```
// prisma/schema.prisma + migrate
npx prisma migrate dev --name init
npx prisma generate
const users = await prisma.user.findMany({ include: { orders: true } });
```

Where to practice

- <https://leetcode.com/problems/meeting-rooms-ii/>

80 LPA Interview Twist

Amazon LP interviews use STAR with 'I' statements and dig 3 levels deep on YOUR specific contribution — team answers score poorly.

Time estimate: 45 minutes

Day 90 – Sat: Tasks CRUD + Comments + Permissions

Problem statement

Full task lifecycle with comments; permissions by team role.

Step-by-step approach (8-hour day)

1. **Hour 1–2:** Task CRUD routes scoped to project; validate project belongs to user's team.
2. **Hour 3:** Status transitions todo → in_progress → done; assignee validation.
3. **Hour 4:** Comments CRUD nested under task.
4. **Hour 5:** Permission matrix: viewer read-only; member create/edit; owner delete project.
5. **Hour 6:** Pagination + filters status/assignee/priority.
6. **Hour 7:** API tests with Supertest skeleton.
7. **Hour 8:** Postman collection export.

Exact Answer

- Follow acceptance criteria in Expected output section
- Implementation matches step-by-step approach requirements

Complete Solution

```
import request from "supertest";
import app from "../src/app";

describe("POST /api/auth/login", () => {
  it("returns 200 with token", async () => {
    const res = await request(app).post("/api/auth/login").send({ email: "test@tes
    expect(res.status).toBe(200);
    expect(res.body.accessToken).toBeDefined();
  });
});
```

Expected output / acceptance criteria

- Non-member gets 403; viewer cannot POST task.

80 LPA Interview Twist

At Google/Meta 80 LPA level, expect follow-up questions on scale (10× traffic), failure modes, and production tradeoffs — not just happy-path implementation.

Time estimate: 8 hours

Day 91 – Sun: Socket.io Real-Time Updates

Problem statement

Broadcast task create/update/delete to team room subscribers.

Step-by-step approach

1. Attach Socket.io to HTTP server; auth middleware on connection via JWT query/header.
2. Join room `team:${teamId}` on subscribe.
3. Emit events: `task:created` , `task:updated` , `task:deleted` .
4. Next.js client hook `useTaskSocket` updates local cache/state.
5. Handle reconnect gracefully.

Exact Answer

- Task CRUD events broadcast to `team:${teamId}` room
- JWT auth on socket connection; client hook updates UI live

Complete Solution

```
io.use((socket, next) => {
  const token = socket.handshake.auth.token;
  try { socket.data.user = verifyAccessToken(token); next(); }
  catch { next(new Error("Unauthorized")); }
});
io.on("connection", (socket) => {
  socket.on("join:team", (teamId) => socket.join(`team:${teamId}`));
});
// On task update: io.to(`team:${teamId}`).emit("task:updated", task);
```

Where to practice

- Socket.io: <https://socket.io/docs/v4/>

80 LPA Interview Twist

Meta/Google security rounds ask about OWASP top 10, token rotation on reuse detection, and zero-trust architecture — auth skeleton must scale to microservices.

Time estimate: 8 hours

Day 92 – Mon: File Upload

Problem statement

Attach files to tasks — Multer local disk or S3 presigned upload.

Step-by-step approach

1. POST `/tasks/:id/attachments` multipart.
2. Store metadata in Mongo; file path or S3 key.
3. Max size 5MB; allowed mime types whitelist.
4. DELETE attachment owner/member only.

Exact Answer

- Follow acceptance criteria in Expected output section
- Implementation matches step-by-step approach requirements

Complete Solution

```
name: CI
on: [pull_request]
jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-node@v4
        with: { node-version: 20 }
      - run: npm ci && npm test && npm run build
```

80 LPA Interview Twist

At Google/Meta 80 LPA level, expect follow-up questions on scale (10× traffic), failure modes, and production tradeoffs — not just happy-path implementation.

Time estimate: 2 hours

Day 93 – Tue: Email Notifications

Problem statement

Send email on task assignment (Nodemailer + Ethereal test or Resend API).

Step-by-step approach

1. Email service abstraction interface.
2. Trigger on assignee change async (don't block response).
3. Template: task title, link to web app.

Exact Answer

- 1B events/day $\approx$ 12K events/sec avg; Kafka partitioned by user_id
- Workers route by user preferences to FCM/SendGrid/Twilio adapters
- Idempotency keys prevent duplicate notifications on retry

Complete Solution

Flow

Event API $\rightarrow$ Kafka (user_id partition) $\rightarrow$ Worker Pool $\rightarrow$ Template Service $\rightarrow$ Provider

Entities

- UserPreferences: { userId, push: bool, email: bool, sms: bool }
- NotificationEvent: { id, userId, type, payload, idempotencyKey }
- Dead letter queue for failed deliveries after 3 retries

80 LPA Interview Twist

At Google/Meta 80 LPA level, expect follow-up questions on scale (10 $\times$ traffic), failure modes, and production tradeoffs — not just happy-path implementation.

Time estimate: 2 hours

Day 94 – Wed: Frontend Dashboard Layout

Problem statement

Next.js dashboard: sidebar teams/projects, header user menu, responsive.

Step-by-step approach

1. App router layout group (`dashboard`).
2. Fetch teams on server or client with auth.
3. Tailwind UI — shadcn optional.

Exact Answer

- Follow acceptance criteria in Expected output section
- Implementation matches step-by-step approach requirements

Complete Solution

```
const session = await stripe.checkout.sessions.create({
  mode: "payment",
  line_items: [{ price: priceId, quantity: 1 }],
  success_url: `${BASE_URL}/success`,
  cancel_url: `${BASE_URL}/cancel`,
});
// Webhook: verify signature, update order status on payment_intent.succeeded
```

80 LPA Interview Twist

Meta/Google security rounds ask about OWASP top 10, token rotation on reuse detection, and zero-trust architecture — auth skeleton must scale to microservices.

Time estimate: 2 hours

Day 95 – Thu: Kanban Task Board UI

Problem statement

Drag-drop or column buttons for status; task cards with assignee, priority, due date.

Step-by-step approach

1. Columns by status; optimistic UI updates.
2. Modal task detail with comments thread.
3. Socket listeners update board live.

Exact Answer

- Follow acceptance criteria in Expected output section
- Implementation matches step-by-step approach requirements

Complete Solution

```
# Vercel (frontend): vercel deploy --prod
# Render/Railway (API): connect repo, set env vars, deploy
# Verify: curl https://api.example.com/health
# Update README with live demo URLs
```

80 LPA Interview Twist

At Google/Meta 80 LPA level, expect follow-up questions on scale (10× traffic), failure modes, and production tradeoffs — not just happy-path implementation.

Time estimate: 2 hours

Day 96 – Fri: LC 322 Coin Change

Where to practice: <https://leetcode.com/problems/coin-change/>

Exact Answer

- `coinChange([1,2,5], 11) → 3` (5+5+1)
- Time: $O(\text{amount} \times \text{coins})$, Space: $O(\text{amount})$

Complete Solution

```
function coinChange(coins, amount) {  
  const dp = Array(amount + 1).fill(Infinity);  
  dp[0] = 0;  
  for (let a = 1; a <= amount; a++) {  
    for (const coin of coins) {  
      if (coin <= a) dp[a] = Math.min(dp[a], dp[a - coin] + 1);  
    }  
  }  
  return dp[amount] === Infinity ? -1 : dp[amount];  
}
```

80 LPA Interview Twist

At Google/Meta 80 LPA level, expect follow-up questions on scale (10× traffic), failure modes, and production tradeoffs — not just happy-path implementation.

Time estimate: 45 minutes

Day 97 – Sat: Docker Compose

Problem statement

`docker-compose up` runs api + mongo + redis + web (web optional dev outside).

Step-by-step approach

1. Dockerfiles multi-stage for API.
2. Compose networks; env files.
3. Volume for mongo data persistence.
4. README "Run with Docker" section.

Exact Answer

- Follow acceptance criteria in Expected output section
- Implementation matches step-by-step approach requirements

Complete Solution

```
function minMeetingRooms(intervals) {
  intervals.sort((a, b) => a[0] - b[0]);
  const ends = [];
  for (const [start, end] of intervals) {
    if (ends.length && ends[0] <= start) ends.shift();
    else ends.push(end);
    ends.sort((a, b) => a - b);
  }
  return ends.length;
}
// Time O(n log n), Space O(n)
```

80 LPA Interview Twist

Netflix/Meta use Redis for rate limiting, pub/sub, and session store — expect distributed Redis cluster and failover discussion.

Time estimate: 8 hours (includes polish)

Day 98 – Sun: Polish + README

Problem statement

Bug bash, architecture diagram, demo GIF, deployment prep.

Expected output / acceptance criteria

- README: features, stack, setup, env vars, API docs link.
- Known issues listed honestly.

Exact Answer

- Follow acceptance criteria in Expected output section
- Implementation matches step-by-step approach requirements

Complete Solution

```
function coinChange(coins, amount) {  
  const dp = Array(amount + 1).fill(Infinity);  
  dp[0] = 0;  
  for (let a = 1; a <= amount; a++)  
    for (const c of coins)  
      if (c <= a) dp[a] = Math.min(dp[a], dp[a - c] + 1);  
  return dp[amount] === Infinity ? -1 : dp[amount];  
}  
// Time O(amount × coins), Space O(amount)
```

80 LPA Interview Twist

Google Cloud SQL interviews cover connection pooling (PgBouncer), read replicas, and EXPLAIN ANALYZE for query optimization at scale.

Time estimate: 8 hours

WEEKS 15-16 – ShopLite + Testing (Days 99-112)

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Stack: Next.js + Express + PostgreSQL (orders) + Redis (cart/session) + Stripe/Razorpay test

Day 99 – Mon: ShopLite Setup

Problem statement

Product catalog schema in PostgreSQL: products, categories, inventory.

Step-by-step approach

1. New repo or monorepo `shoplite`.
2. Prisma models Product, Category, ProductImage optional.
3. Admin seed products.

Exact Answer

- PostgreSQL schema: Product, Category, inventory tracking
- Prisma models with seed data for catalog

Complete Solution

```
model Product {
  id          String    @id @default(uuid())
  name        String
  price       Decimal   @db.Decimal(10, 2)
  stock       Int       @default(0)
  categoryId  String
  category    Category @relation(fields: [categoryId], references: [id])
}
```

80 LPA Interview Twist

Google Cloud SQL interviews cover connection pooling (PgBouncer), read replicas, and EXPLAIN ANALYZE for query optimization at scale.

Time estimate: 2 hours

Day 100 – Tue: Catalog API + Admin Routes

Problem statement

Public GET products; admin POST/PUT/DELETE with RBAC.

Step-by-step approach

1. Query filters: category, price range, search q.
2. Admin middleware on mutate routes.
3. Pagination consistent with TaskFlow patterns.

Exact Answer

- Public GET with filters (category, price, search q); admin POST/PUT/DELETE requires admin role
- Pagination: `{ data, total, page, pages }`

Complete Solution

```
router.get("/products", async (req, res) => {
  const { category, q, minPrice, maxPrice, page = 1, limit = 20 } = req.query;
  const where = {
    ...(category && { category: String(category) }),
    ...(q && { name: { contains: String(q), mode: "insensitive" } }),
    ...(minPrice && { price: { gte: Number(minPrice) } }),
    ...(maxPrice && { price: { lte: Number(maxPrice) } }),
  };
  const [data, total] = await Promise.all([
    prisma.product.findMany({ where, skip: (Number(page)-1)*Number(limit), take: Number(limit) }),
    prisma.product.count({ where })
  ]);
  res.json({ data, total, page: Number(page), pages: Math.ceil(total / Number(limit)) });
  router.post("/products", authenticate, authorize("admin"), createProduct);
});
```

80 LPA Interview Twist

Flagship project reviews at 80 LPA focus on real-time consistency, conflict resolution, and horizontal scaling of Socket.io with Redis adapter.

Time estimate: 2 hours

Day 101 – Wed: Cart + Redis Session

Problem statement

Guest and logged-in cart stored in Redis; merge on login.

Step-by-step approach

1. Session id cookie for guest cart key `cart:${sessionId}`.
2. Cart JSON: `[{ productId, qty }]`.
3. Validate stock on add.

Exact Answer

- Guest cart stored at `cart:${sessionId}` in Redis as JSON array
- Login merges guest cart into user cart; stock validated on add

Complete Solution

```
async function addToCart(sessionId: string, productId: string, qty: number) {
  const key = `cart:${sessionId}`;
  const cart: { productId: string; qty: number }[] = JSON.parse(await redis.get(key));
  const product = await prisma.product.findUnique({ where: { id: productId } });
  if (!product || product.stock < qty) throw new AppError(400, "Insufficient stock");
  const existing = cart.find(i => i.productId === productId);
  if (existing) existing.qty += qty; else cart.push({ productId, qty });
  await redis.setex(key, 86400, JSON.stringify(cart));
}
```

80 LPA Interview Twist

Netflix/Meta use Redis for rate limiting, pub/sub, and session store — expect distributed Redis cluster and failover discussion.

Time estimate: 2 hours

Day 102 – Thu: Checkout + Order Creation

Problem statement

Transactional order create: decrement inventory, clear cart, order row + line items.

Step-by-step approach

1. Prisma `$transaction` for order + items + stock.
2. Order statuses: pending → paid → shipped.

3. Idempotency key header optional advanced.

Exact Answer

- Prisma `$transaction` creates order, line items, decrements stock atomically
- Cart cleared on success; rollback on any failure

Complete Solution

```
await prisma.$transaction(async (tx) => {
  const order = await tx.order.create({ data: { userId, total, status: "pending" } })
  for (const item of cartItems) {
    const product = await tx.product.update({
      where: { id: item.productId },
      data: { stock: { decrement: item.qty } },
    });
    if (product.stock < 0) throw new Error("Insufficient stock");
    await tx.orderItem.create({ data: { orderId: order.id, productId: item.productId } })
  }
  await redis.del(`cart:${sessionId}`);
});
```

80 LPA Interview Twist

At Google/Meta 80 LPA level, expect follow-up questions on scale (10× traffic), failure modes, and production tradeoffs — not just happy-path implementation.

Time estimate: 2 hours

Day 103 – Fri: LC 300 LIS

Where to practice: <https://leetcode.com/problems/longest-increasing-subsequence/>

Exact Answer

- Optimal solution with stated time/space complexity
- Passes all LeetCode test cases including edge cases

Complete Solution

```
describe("calculateTotal", () => {  
  it("sums cart items", () => {  
    expect(calculateTotal([  
      { price: 10, qty: 2 },  
      { price: 5, qty: 1 }  
    ])).toBe(25)  
  });  
});
```

80 LPA Interview Twist

At Google/Meta 80 LPA level, expect follow-up questions on scale (10× traffic), failure modes, and production tradeoffs — not just happy-path implementation.

Time estimate: 60 minutes

Day 104 – Sat: Payment Integration

Problem statement

Stripe or Razorpay test mode checkout session; webhook marks order paid.

Step-by-step approach

1. Create checkout session endpoint.
2. Webhook signature verification.
3. Update order on `payment_intent.succeeded`.

Exact Answer

- All acceptance criteria from Expected output section met
- Implementation follows step-by-step approach; tested and documented in README

Complete Solution

```
// Follow the step-by-step approach documented above for this exercise.  
// Key deliverables: working code, tests, README documentation.  
// Verify with: npm test && curl/httpie manual testing
```

Where to practice

- Stripe test: <https://stripe.com/docs/testing>

80 LPA Interview Twist

At Google/Meta 80 LPA level, expect follow-up questions on scale (10× traffic), failure modes, and production tradeoffs — not just happy-path implementation.

Time estimate: 8 hours

Day 105 – Sun: Order Management + Admin Dashboard

Problem statement

Admin view orders; user order history page.

Exact Answer

- All acceptance criteria from Expected output section met
- Implementation follows step-by-step approach; tested and documented in README

Complete Solution

```
// Follow the step-by-step approach documented above for this exercise.  
// Key deliverables: working code, tests, README documentation.  
// Verify with: npm test && curl/httpie manual testing
```

80 LPA Interview Twist

At Google/Meta 80 LPA level, expect follow-up questions on scale (10× traffic), failure modes, and production tradeoffs — not just happy-path implementation.

Time estimate: 8 hours

Day 106 – Mon: Jest Unit Tests

Problem statement

Unit test pure functions: pricing, cart merge, slug helpers.

Step-by-step approach

1. Jest config for API package.
2. ≥80% coverage on utils target optional.
3. Mock dates and IDs deterministically.

Exact Answer

- All acceptance criteria from Expected output section met
- Implementation follows step-by-step approach; tested and documented in README

Complete Solution

```
// Follow the step-by-step approach documented above for this exercise.  
// Key deliverables: working code, tests, README documentation.  
// Verify with: npm test && curl/httpie manual testing
```

Where to practice

- Jest: <https://jestjs.io/docs/getting-started>

80 LPA Interview Twist

Amazon LP interviews use STAR with 'I' statements and dig 3 levels deep on YOUR specific contribution — team answers score poorly.

Time estimate: 2 hours

Day 107 – Tue: Supertest API Tests

Problem statement

Integration tests for auth, products, checkout happy path.

Step-by-step approach

1. Spin test DB or in-memory mongo substitute per test file strategy.
2. `request(app).post('/api/auth/login')...`
3. CI must run tests headless.

Exact Answer

- All acceptance criteria from Expected output section met
- Implementation follows step-by-step approach; tested and documented in README

Complete Solution

```
// Follow the step-by-step approach documented above for this exercise.  
// Key deliverables: working code, tests, README documentation.  
// Verify with: npm test && curl/httpie manual testing
```

80 LPA Interview Twist

Meta/Google security rounds ask about OWASP top 10, token rotation on reuse detection, and zero-trust architecture — auth skeleton must scale to microservices.

Time estimate: 2 hours

Day 108 – Wed: React Testing Library

Problem statement

Component tests: product card renders, add to cart button calls handler.

Where to practice

- RTL: <https://testing-library.com/docs/react-testing-library/intro/>

Exact Answer

- All acceptance criteria from Expected output section met
- Implementation follows step-by-step approach; tested and documented in README

Complete Solution

```
// Follow the step-by-step approach documented above for this exercise.  
// Key deliverables: working code, tests, README documentation.  
// Verify with: npm test && curl/httpie manual testing
```

80 LPA Interview Twist

At Google/Meta 80 LPA level, expect follow-up questions on scale (10× traffic), failure modes, and production tradeoffs — not just happy-path implementation.

Time estimate: 2 hours

Day 109 – Thu: GitHub Actions CI

Problem statement

Workflow: lint, test, build on PR to main.

Step-by-step approach

1. `.github/workflows/ci.yml`.
2. Matrix node 20; cache npm.
3. Fail PR if tests fail.

Exact Answer

- All acceptance criteria from Expected output section met
- Implementation follows step-by-step approach; tested and documented in README

Complete Solution

```
// Follow the step-by-step approach documented above for this exercise.  
// Key deliverables: working code, tests, README documentation.  
// Verify with: npm test && curl/httpie manual testing
```

Where to practice

- GitHub Actions: <https://docs.github.com/en/actions>

80 LPA Interview Twist

At Google/Meta 80 LPA level, expect follow-up questions on scale (10× traffic), failure modes, and production tradeoffs — not just happy-path implementation.

Time estimate: 2 hours

Day 110 – Fri: LC 347 Top K Frequent

Where to practice: <https://leetcode.com/problems/top-k-frequent-elements/>

Exact Answer

- Optimal solution with stated time/space complexity
- Passes all LeetCode test cases including edge cases

Complete Solution

```
import request from "supertest";  
import app from "../src/app";  
  
describe("POST /api/auth/login", () => {  
  it("returns 200 with token", async () => {  
    const res = await request(app).post("/api/auth/login").send({ email: "test@tes  
    expect(res.status).toBe(200);  
    expect(res.body.accessToken).toBeDefined();  
  });  
});
```

80 LPA Interview Twist

At Google/Meta 80 LPA level, expect follow-up questions on scale (10× traffic), failure modes, and production tradeoffs — not just happy-path implementation.

Time estimate: 45 minutes

Day 111 – Sat: Deploy Both Projects

Problem statement

TaskFlow + ShopLite live: Vercel frontend, Render/Railway API, Atlas + Neon DBs.

Step-by-step approach

1. Environment variables on host platforms.
2. CORS production origins.
3. Health checks for Render.
4. Smoke test production URLs.

Exact Answer

- All acceptance criteria met per step-by-step approach
- Code committed to repo with README documentation

Complete Solution

```
// Follow the 8-step build approach documented above
// Commit each step separately with meaningful messages
// Document architecture decisions in README
```

Expected output / acceptance criteria

- Live demo URLs in README and resume.

80 LPA Interview Twist

Flagship project reviews at 80 LPA focus on real-time consistency, conflict resolution, and horizontal scaling of Socket.io with Redis adapter.

Time estimate: 8 hours

Day 112 – Sun: Architecture Diagrams + Resume Bullets

Problem statement

Create diagrams (draw.io or Mermaid); write 3 resume bullets per project with metrics.

Step-by-step approach

1. C4 context + container diagrams.
2. Bullets: tech, scale, outcome ("Reduced load time 40% via Redis cache").

3. 150+ LeetCode milestone self-check.

Exact Answer

- All acceptance criteria from Expected output section met
- Implementation follows step-by-step approach; tested and documented in README

Complete Solution

```
// Follow the step-by-step approach documented above for this exercise.  
// Key deliverables: working code, tests, README documentation.  
// Verify with: npm test && curl/httpie manual testing
```

Expected output / acceptance criteria

- PDF or PNG diagrams in repos; resume updated.

80 LPA Interview Twist

Netflix/Meta use Redis for rate limiting, pub/sub, and session store — expect distributed Redis cluster and failover discussion.

Time estimate: 8 hours

Month 4 LeetCode Reference

LC 253, 268, 283, 287, 300, 322, 347, 349, 371, 373, 378, 392, 399, 416, 424, 435 — apply standard approach template from Week 1.

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---## Month 4 Completion Checklist

Milestone	Done
TaskFlow feature-complete	☐
Socket.io real-time	☐
Docker Compose	☐
ShopLite checkout + payment test	☐
CI pipeline green	☐
Both projects deployed	☐
150+ LeetCode	☐

Total estimated time: ~112 hours (Days 85–112)